Highlights

Parallelizing Quantized LLM Inference Under a Fixed Numerical Reference

- Physical KV ranges and arithmetic groups impose different parallelism constraints.
- Local attention cuts retain wider work while preserving a fixed numerical reference.
- Output ownership and state obligations connect compatible execution to its consumer.
- A strong 32B control isolates a 1.51% target-only resident-service reduction.
- Boundary geometry and extra speculative work explain the observed performance limits.